Lab 5: Analysis of Variance

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Setup

1. Coastal Giant Salamander

```
# Filter for cascades (C), pools (P), and side channels (SC) for only Coastal giant salamander species.
str(and_vertebrates)
```

```
## tibble [32,209 x 16] (S3: tbl_df/tbl/data.frame)
##  $ year       : num [1:32209] 1987 1987 1987 1987 1987 ...
##  $ sitecode   : chr [1:32209] "MACKCC-L" "MACKCC-L" "MACKCC-L" "MACKCC-L" ...
##  $ section    : chr [1:32209] "CC" "CC" "CC" "CC" ...
##  $ reach      : chr [1:32209] "L" "L" "L" "L" ...
##  $ pass       : num [1:32209] 1 1 1 1 1 1 1 1 1 1 ...
##  $ unitnum    : num [1:32209] 1 1 1 1 1 1 1 1 1 1 ...
##  $ unittype   : chr [1:32209] "R" "R" "R" "R" ...
##  $ vert_index : num [1:32209] 1 2 3 4 5 6 7 8 9 10 ...
##  $ pitnumber  : num [1:32209] NA NA NA NA NA NA NA NA NA NA ...
##  $ species    : chr [1:32209] "Cutthroat trout" "Cutthroat trout" "Cutthroat trout" "Cutthroat trout
##  $ length_1_mm: num [1:32209] 58 61 89 58 93 86 107 131 103 117 ...
##  $ length_2_mm: num [1:32209] NA NA NA NA NA NA NA NA NA NA ...
##  $ weight_g   : num [1:32209] 1.75 1.95 5.6 2.15 6.9 5.9 10.5 20.6 9.55 13 ...
##  $ clip       : chr [1:32209] "NONE" "NONE" "NONE" "NONE" ...
##  $ sampledate : Date[1:32209], format: "1987-10-07" "1987-10-07" ...
##  $ notes      : chr [1:32209] NA NA NA NA ...
```

```
view(and_vertebrates)

salamander_data <- and_vertebrates %>%
  filter(unittype %in% c("C", "P", "SC"),
         species == "Coastal giant salamander")

salamander_fig1 <- ggplot(salamander_data,
                          aes(x = length_2_mm, fill = unittype)) +
  geom_histogram() +
    facet_wrap(~unittype) +
  scale_fill_manual(values = c("C" = "darkorange",
                               "P" = "purple",
                               "SC" = "cyan4")) +
```

```r
  labs(x = "Snout-to-Tail Length (mm)",
       y = "Salamander Count",
       title = "Distribution of Coastal Giant Salamander Lengths Across Various
       Habitat Types",
       caption = "Figure 1. Coastal Giant Salamanders in cascades (C), pools (P),
       and side channel (SC) habitats. Dataset collected by H.J. Andrews Long
       Term Ecological Research program within the Mack Creek watershed in Oregon") +
  theme(legend.position = "none")

salamander_fig1
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

## Warning: Removed 220 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

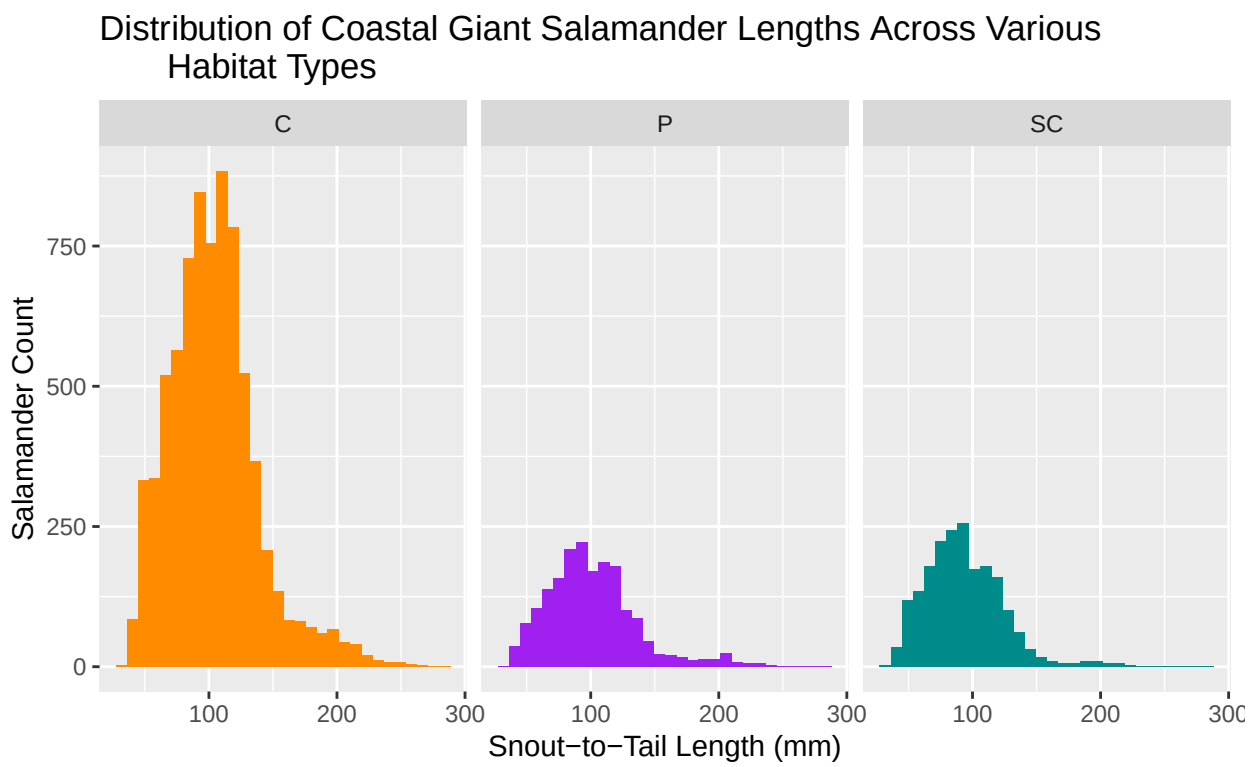


Figure 1. Coastal Giant Salamanders in cascades (C), pools (P), and side channel (SC) habitats. Dataset collected by H.J. Andrews Long Term Ecological Research program within the Mack Creek watershed in Oregon

```r
Habitat_C <- salamander_data %>%
  filter(unittype == "C")
skew_C <- skewness(Habitat_C$length_2_mm, na.rm = TRUE)
kurt_C <- kurtosis(Habitat_C$length_2_mm, na.rm = TRUE)

Habitat_P <- salamander_data %>%
  filter(unittype == "P")
skew_P <- skewness(Habitat_P$length_2_mm, na.rm = TRUE)
```

```r
kurt_P <- kurtosis(Habitat_P$length_2_mm, na.rm = TRUE)

Habitat_SC <- salamander_data %>%
  filter(unittype == "SC")
skew_SC <- skewness(Habitat_SC$length_2_mm, na.rm = TRUE)
kurt_SC <- kurtosis(Habitat_SC$length_2_mm, na.rm = TRUE)

salamander_data_log <- salamander_data %>%
  mutate(length_log10 = log10(length_2_mm))

Habitat_C <- salamander_data_log %>%
  filter(unittype == "C")
skew_C_log <- skewness(Habitat_C$length_log10, na.rm = TRUE)
kurt_C_log <- kurtosis(Habitat_C$length_log10, na.rm = TRUE)

Habitat_P <- salamander_data_log %>%
  filter(unittype == "P")
skew_P_log <- skewness(Habitat_P$length_log10, na.rm = TRUE)
kurt_P_log <- kurtosis(Habitat_P$length_log10, na.rm = TRUE)

Habitat_SC <- salamander_data_log %>%
  filter(unittype == "SC")
skew_SC_log <- skewness(Habitat_SC$length_log10, na.rm = TRUE)
kurt_SC_log <- kurtosis(Habitat_SC$length_log10, na.rm = TRUE)
```

```r
salamander_var <- bartlett.test(salamander_data_log$length_log10,
                                salamander_data_log$unittype)
salamander_var
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  salamander_data_log$length_log10 and salamander_data_log$unittype
## Bartlett's K-squared = 11.213, df = 2, p-value = 0.003674
```

```r
salamander_ANOVA <- aov(length_log10 ~ unittype, data = salamander_data_log)

summary(salamander_ANOVA)
```

```
##                Df Sum Sq Mean Sq F value Pr(>F)
## unittype        2    3.2  1.5993   76.35 <2e-16 ***
## Residuals   11411  239.0  0.0209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 220 observations deleted due to missingness
```

```r
salamander_Tukey <- TukeyHSD(salamander_ANOVA)
salamander_Tukey
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
```

```
##
## Fit: aov(formula = length_log10 ~ unittype, data = salamander_data_log)
##
## $unittype
##             diff         lwr          upr     p adj
## P-C  -0.01310767 -0.02187368 -0.004341656 0.0013319
## SC-C -0.04502145 -0.05359132 -0.036451579 0.0000000
## SC-P -0.03191378 -0.04286293 -0.020964629 0.0000000
```

Question 1 Answers 1a. All habitats are not normally distribution with data strongly skewed to the right (0.5 > skewness value > 1.0, meaning moderately skewed) with heavy tails (kurtosis value > 3, meaning leptokurtic), which is displayed in Figure 1. After applying the log-transformation to the snout-to-length data, quick skewness and kurtosis values indicate that all three habitats closely meet the criteria for normal distribution (skewness around 0 AND kurtosis around 3 values).

1b. Reject the null hypothesis of Bartlett's test that there is equal variances across groups because the p-value (0.003674) is much smaller than 0.05. In other words, variances are not homogeneous after applying the log-transformation for salamander length and habitat type.

1c. Data from part b did not display approximately equal variances!

2. Coastal Giant Salamander

```r
# a. Filter for reach and cutthroat trout species. Plot and test whether or not there is normal distrib

trout_data <- and_vertebrates %>%
  filter(species == "Cutthroat trout")
unique(trout_data$reach)
```

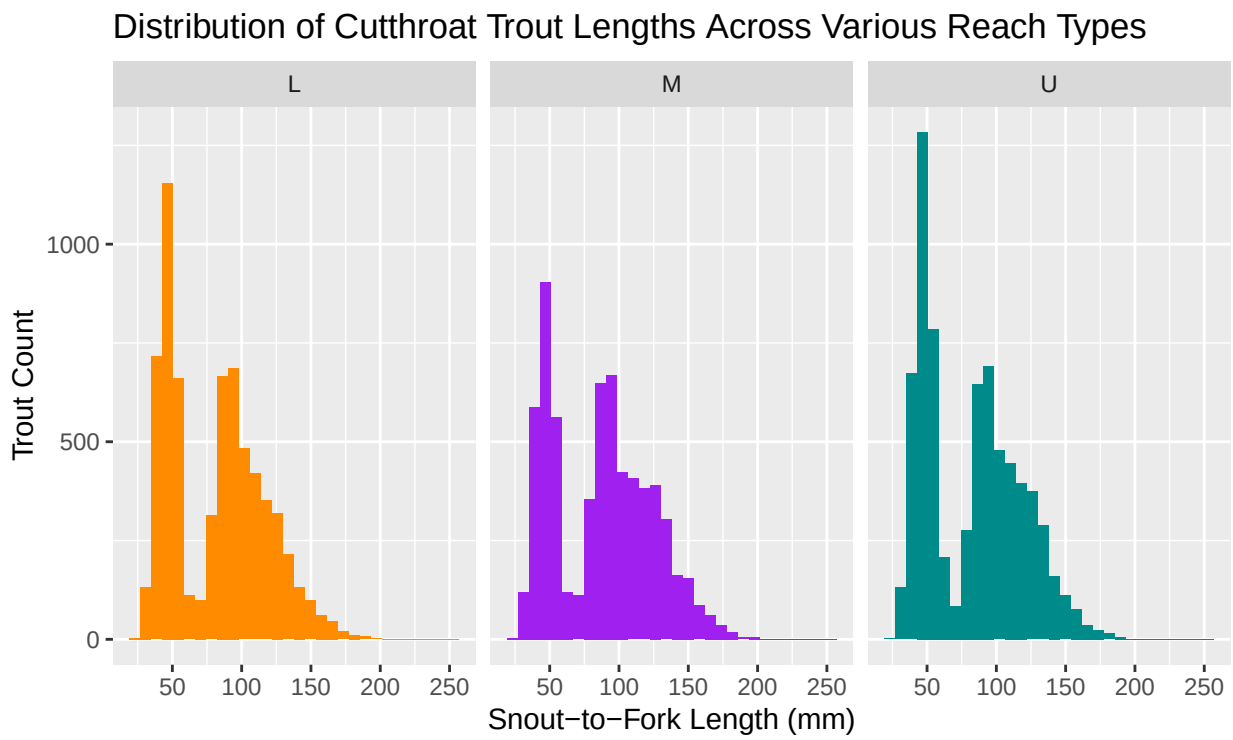
```
## [1] "L" "M" "U"
```

```r
trout_fig2 <- ggplot(trout_data,
                         aes(x = length_1_mm, fill = reach)) +
  geom_histogram() +
    facet_wrap(~reach) +
  scale_fill_manual(values = c("L" = "darkorange",
                               "M" = "purple",
                               "U" = "cyan4")) +
  labs(x = "Snout-to-Fork Length (mm)",
       y = "Trout Count",
       title = "Distribution of Cutthroat Trout Lengths Across Various Reach Types",
       caption = "Cutthroat Trout in Mack Creek Cut Forest lower reach (L),
       middle reach (M), and upper reach (U) types. Dataset collected by H.J.
       Andrews Long Term Ecological Research program within the Mack Creek watershed
       in Oregon") +
  theme(legend.position = "none")

trout_fig2
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Cutthroat Trout in Mack Creek Cut Forest lower reach (L), middle reach (M), and upper reach (U) types. Dataset collected by H.J. Andrews Long Term Ecological Research program within the Mack Creek watershed in Oregon

```
Reach_L <- trout_data %>%
  filter(reach == "L")
skew_L <- skewness(Reach_L$length_1_mm, na.rm = TRUE)
kurt_L <- kurtosis(Reach_L$length_1_mm, na.rm = TRUE)

Reach_M <- trout_data %>%
  filter(reach == "M")
skew_M <- skewness(Reach_M$length_1_mm, na.rm = TRUE)
kurt_M <- kurtosis(Reach_M$length_1_mm, na.rm = TRUE)

Reach_U <- trout_data %>%
  filter(reach == "U")
skew_U <- skewness(Reach_U$length_1_mm, na.rm = TRUE)
kurt_U <- kurtosis(Reach_U$length_1_mm, na.rm = TRUE)
```

```
trout_var <- bartlett.test(trout_data$length_1_mm,
                           trout_data$reach)
trout_var
```

```
##
##  Bartlett test of homogeneity of variances
##
```

```
## data:  trout_data$length_1_mm and trout_data$reach
## Bartlett's K-squared = 10.697, df = 2, p-value = 0.004756
```

```r
trout_ANOVA <- aov(length_1_mm ~ reach, data = trout_data)

summary(trout_ANOVA)
```

```
##                Df   Sum Sq Mean Sq F value Pr(>F)
## reach           2   116788   58394   46.87 <2e-16 ***
## Residuals   20425 25447121    1246
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 5 observations deleted due to missingness
```

```r
trout_Tukey <- TukeyHSD(trout_ANOVA)
trout_Tukey
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = length_1_mm ~ reach, data = trout_data)
##
## $reach
##           diff       lwr       upr     p adj
## M-L  5.4089692  3.970386  6.847553 0.0000000
## U-L  0.5924704 -0.811419  1.996360 0.5837238
## U-M -4.8164988 -6.231043 -3.401955 0.0000000
```

```r
trout_means <- trout_data %>%
  group_by(reach) %>%
  summarise(mean_length = mean(length_1_mm, na.rm = TRUE))

trout_means
```

```
## # A tibble: 3 x 2
##   reach mean_length
##   <chr>       <dbl>
## 1 L            81.6
## 2 M            87.0
## 3 U            82.2
```

```r
trout_fig3 <- ggplot(trout_means,
                     aes(x = reach, y = mean_length, fill = reach)) +
  geom_col() +
  scale_fill_manual(values = c("L" = "darkorange",
                               "M" = "purple",
                               "U" = "cyan4")) +
  labs(x = "Reach Type",
       y = "Mean Snout-to-Fork Length (mm)",
       title = "Mean Cutthroat Trout Lengths Across Stream Reaches",
       caption = "Figure 3. Mean Cutthroat Trout lengths in Mack Creek Cut
```

```
      Forest lower (L), middle (M), and upper (U) reaches. Dataset collected
      by the H.J. Andrews Long Term Ecological Research program.") +
  theme(legend.position = "none")

trout_fig3
```

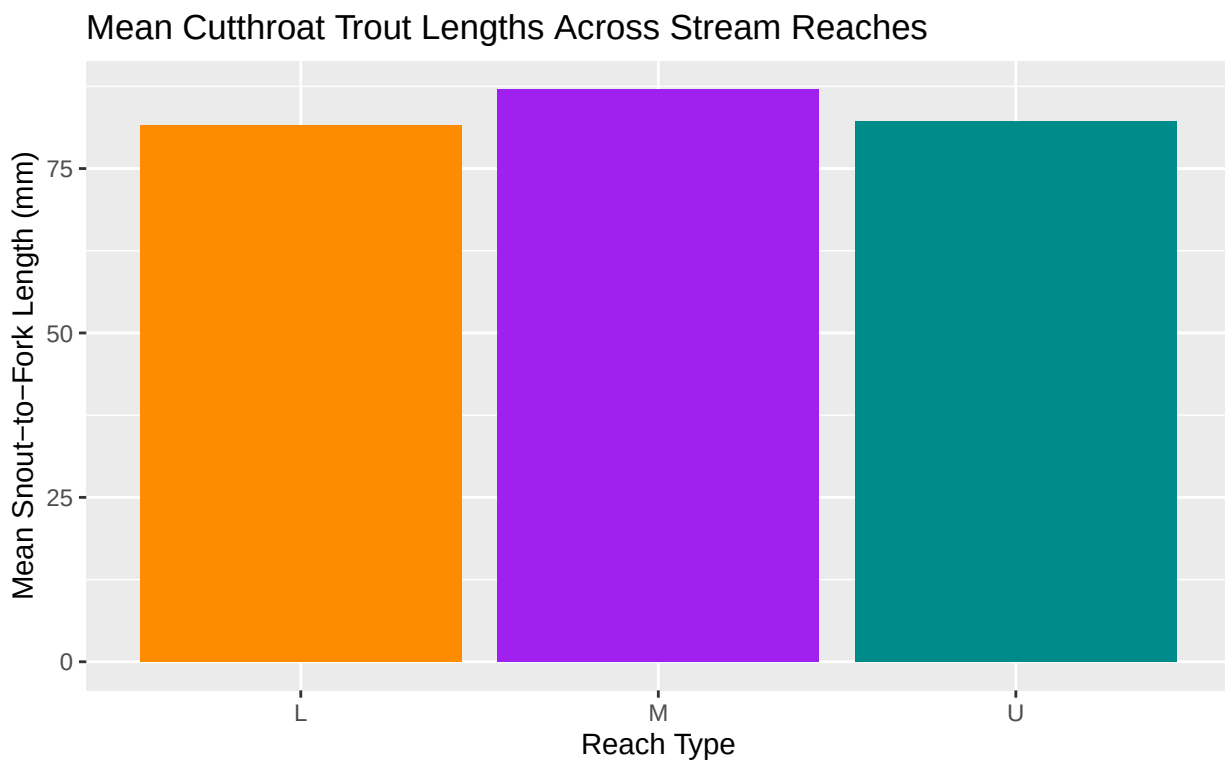


Figure 3. Mean Cutthroat Trout lengths in Mack Creek Cut Forest lower (L), middle (M), and upper (U) reaches. Dataset collected by the H.J. Andrews Long Term Ecological Research program.

Question 2 Answers 2a. All reaches are meet the criteria for normal distribution with data slightly skewed to the right (0.4 > skewness value > 0, meaning moderately skewed yet closer to almost normal distribution) with very light tails (kurtosis value < 3, meaning slightly platykurtic), which is displayed in Figure 2.

2b. Reject the null hypothesis of Bartlett's test that there is equal variances across groups because the p-value (0.004756) is much smaller than 0.05. In other words, variances in trout length are not homogeneous across different reaches.

2c. There is evidence of significant differences between trout length and reach type because the p-value (2e-16) is smaller than 0.05 and a large F-statistic (46.87; meaning variation between reaches are large and variation within reaches are small). Post-hoc Tukey comparisons result with trout lengths similar between Uupper and Lower reaches (p = 0.584, approximately 0.59 mm differences in values). Middle reach were significantly larger than those from both U and L reaches (p < 0.000001 for both comparisons), meaning that Middle reach Cutthroat troat were generally larger in length.